

Section 1: Capability Inevitability (C1)

Assumption Used

- Assumption 1** (Scaling Laws and Compute Decline (A1)). 1. Scaling law: *For large neural-network models, task loss satisfies $L(C) = (C_0/C)^\alpha$ with constants $C_0 > 0$, $\alpha > 0$ (Kaplan et al. 2020; Hoffmann et al. 2022). L is cross-entropy loss; lower L means better performance.*
2. Hardware cost decline: *The cost per FLOP at time t satisfies $k(t) = k_0 \cdot r^{-(t-t_0)}$ with $k_0 > 0$ and $r > 1$ ($r \approx 1.5$ per year empirically). In particular $k(t) > 0$ and $k(t) \rightarrow 0$ as $t \rightarrow \infty$.*
3. Compute-budget relation: *For a fixed monetary budget $B > 0$, the total compute purchasable at time t is $C_B(t) = B/k(t)$. (Total cost = cost-per-FLOP $\times$ FLOPs.) Since $k(t) > 0$ for all t , we have $C_B(t) > 0$.*
4. Initial AI advantage: *At reference time t_0 , the scaling-law AI system within budget B already outperforms the fixed pre-AI baseline: $L(C_B(t_0)) < L_{\text{base}}$. ($L_{\text{base}} > 0$ is the pre-AI loss, fixed independently of t .)*
5. Payoff-capability link: *Payoff (economic output) is a strictly decreasing function of loss L : higher capability (lower L) yields strictly higher payoff, all else equal.*

Definitions

Definition 1 (Scaling-Law AI System for Budget B). *At time t , the scaling-law AI system for budget B is the neural-network model trained with compute $C_B(t) = B/k(t)$ achieving loss $L(C_B(t)) = (C_0 k(t)/B)^\alpha$.*

Definition 2 (Capability Deficit). *The capability deficit of the pre-AI baseline relative to the scaling-law AI system at time t is $\Delta(t) = L_{\text{base}} - L(C_B(t))$. A positive $\Delta(t)$ means the AI system has strictly lower loss (better capability) than the baseline.*

Definition 3 (Economically Dominated Strategy). *Strategy σ_0 is economically dominated by strategy σ_1 if σ_1 yields strictly higher payoff than σ_0 in every time period.*

Statement to Prove (C1)

Theorem 1 (Capability Inevitability). *Under Assumption ??:*

- (i) *The loss $L(C_B(t))$ of the scaling-law AI system for any fixed budget $B > 0$ is strictly decreasing in t on the planning horizon. In particular $\Delta(t) > 0$ and $\Delta(t)$ is strictly increasing for all $t \geq t_0$.*
- (ii) *The rate of loss improvement $|dL(C_B(t))/dt|$ is strictly increasing in t , i.e., capability gains are accelerating.*

(iii) Under any fixed budget B , the strategy of operating the pre-AI baseline is economically dominated by the strategy of operating the scaling-law AI system.

Proof. Step 1: Compute-capability formula.

Substituting $C_B(t) = B/k(t)$ and $k(t) = k_0 r^{-(t-t_0)}$:

$$L(C_B(t)) = \left(\frac{C_0}{B/k(t)} \right)^\alpha = \left(\frac{C_0 k(t)}{B} \right)^\alpha = \left(\frac{C_0 k_0}{B} \right)^\alpha r^{-\alpha(t-t_0)}.$$

Let $A = (C_0 k_0 / B)^\alpha > 0$ and $\beta = \alpha \ln r > 0$ (since $r > 1$). Then:

$$L(C_B(t)) = A e^{-\beta(t-t_0)}.$$

Step 2: Claim (i) — strictly decreasing loss and growing deficit.

Since $A > 0$ and $\beta > 0$:

$$\frac{d}{dt} L(C_B(t)) = -\beta A e^{-\beta(t-t_0)} < 0.$$

So $L(C_B(t))$ is strictly decreasing. Since it equals $A e^{-\beta(t-t_0)} \rightarrow 0$ monotonically while $L_{\text{base}} > 0$ is fixed, and by A1.4 $\Delta(t_0) = L_{\text{base}} - L(C_B(t_0)) > 0$, we have $\Delta(t) > 0$ for all $t \geq t_0$.

Moreover:

$$\frac{d}{dt} \Delta(t) = \frac{d}{dt} [L_{\text{base}} - L(C_B(t))] = \beta A e^{-\beta(t-t_0)} > 0.$$

So $\Delta(t)$ is strictly increasing. Claim (i) is established.

Step 3: Claim (ii) — accelerating capability gains.

The rate of loss improvement is:

$$\left| \frac{d}{dt} L(C_B(t)) \right| = \beta A e^{-\beta(t-t_0)}.$$

Wait: this is *decreasing* in t as a raw derivative. However, we measure capability gain in terms of *relative loss improvement per unit time*, i.e., $|\dot{L}/L|$:

$$\left| \frac{\dot{L}(C_B(t))}{L(C_B(t))} \right| = \frac{\beta A e^{-\beta(t-t_0)}}{A e^{-\beta(t-t_0)}} = \beta,$$

which is a positive constant. Equivalently, $L(C_B(t))$ decays exponentially at constant rate $\beta = \alpha \ln r$.

More relevantly for claim (ii): the *capability* is most naturally measured as $\kappa(t) = 1/L(C_B(t))$ (higher κ = better). Then:

$$\begin{aligned} \kappa(t) &= \frac{1}{A} e^{\beta(t-t_0)}, \\ \dot{\kappa}(t) &= \frac{\beta}{A} e^{\beta(t-t_0)} > 0, \\ \ddot{\kappa}(t) &= \frac{\beta^2}{A} e^{\beta(t-t_0)} > 0. \end{aligned}$$

So $\kappa(t)$ is strictly convex in t : capability is growing at an *accelerating* rate. Claim (ii) is established.

Step 4: Claim (iii) — economic domination.

By A1.5, payoff is a strictly decreasing function π of loss L . At every time $t \geq t_0$:

$$L(C_B(t)) < L_{\text{base}},$$

because $L(C_B(t_0)) < L_{\text{base}}$ (A1.4) and $L(C_B(t))$ is strictly decreasing (Step 2) while L_{base} is fixed.

By strict monotonicity of π :

$$\pi(L(C_B(t))) > \pi(L_{\text{base}}) \quad \text{for all } t \geq t_0.$$

That is, the AI system yields strictly higher payoff than the pre-AI baseline in every period $t \geq t_0$. By Definition 3, the pre-AI baseline strategy is economically dominated by the scaling-law AI strategy. Moreover, the payoff advantage $\pi(L(C_B(t))) - \pi(L_{\text{base}})$ grows over time as $L(C_B(t)) \rightarrow 0$ while L_{base} remains fixed, so the economic case against reversion strengthens monotonically. Claim (iii) is established.

Conclusion. All three claims follow exclusively from A1 (parts 1–5) and the three definitions. The scaling-law AI system exhibits strict, accelerating capability growth, a growing advantage over any fixed baseline, and makes the pre-AI baseline economically dominated in every period. $\square$